FareShare

Verified College Ride-Pooling & Fair Distance-Split Platform

MVP MILESTONE SUBMISSION

HACKATHON EVENT	ENGINEERING TEAM	WORKING PROTOTYPE STATUS
coNStruCT 2.0 Product × AI (NST & The Product Folks)	Team EnderMen (5 Builders)	Production Deployed (Vite 6 / React 19)

75%	< 50ms	4 Corridors	1-Click
SOLO FARE REDUCTION	ZERO-LATENCY MATCHING	AIRPORT, STATION & ISBT	DIRECT UPI SPLIT

1. What We Have Built (Production Prototype Overview)

Rather than delivering static wireframes or non-interactive mockups, Team EnderMen engineered and deployed a fully functional, end-to-end web platform. Users can execute complete student ride-pooling journeys in real time—from bilingual route discovery to real-time seat decrementing and pre-filled UPI settlement.

Campus-Verified Safe Circles & Filters

Features a dedicated **"Girls-Only Safe Pools"** toggle ensuring female students travel exclusively with verified female peers. Incorporates **Luggage Allowance matching** (Light Backpack, 1 Trolley, Heavy Boot Space) so passengers never arrive to an overloaded cab.

Bilingual NLP Intent Parser (Voice & Text)

Integrated with the browser **Web Speech API** and an in-browser algorithmic parser supporting English, Hindi, and Hinglish queries (e.g. *"kal shaam IGI T3 jana hai"*). Auto-extracts departure time, destination, and urgency tags without cloud latency.

Multi-Modal Solo Benchmark Strips

Every pool card features interactive comparison pills linking directly to Uber Go, Ola Mini, and the Delhi Metro (DMRC) official route & fare calculator, allowing users to verify prevailing solo rates with zero login barriers.

1-Click UPI Settlement & Ephemeral Privacy

In-app group ride drawers embed pre-filled **NPCI Universal UPI deep-links** (GPay, PhonePe, Paytm) carrying the exact per-person fare share. Group coordination automatically purges post-ride to protect student privacy.

2. Real-Time Economic Impact (Solo vs. FareShare 4-Way Pool)

CAMPUS TRANSIT CORRIDOR	SOLO UBER GO	SOLO OLA MINI	DELHI METRO (DMRC)	FARESHARE 4-WAY POOL	STUDENT SAVINGS
Campus ⇄ IGI Airport T3 (44 KM • ~65m)	₹1,400	₹1,350	₹60 (80m, 2 switches)	**₹350 / rider**	Save ₹1,050 (75%)
Campus ⇄ NDLS Railway Station (31 KM • ~50m)	₹900	₹880	₹50 (65m, Yellow line)	**₹225 / rider**	Save ₹675 (75%)
Campus ⇄ Anand Vihar ISBT (28 KM • ~45m)	₹800	₹780	₹50 (60m, Blue line)	**₹200 / rider**	Save ₹600 (75%)
Campus ⇄ Cyber City / Hub (14 KM • ~25m)	₹450	₹430	₹30 (40m, Rapid connector)	**₹113 / rider**	Save ₹337 (75%)

FareShare • Wireframes & UI/UX Design System

Information Architecture, Mobile Viewports & Human-Centric Ergonomics

DEDICATED WIREFRAMES

CORE SYSTEM MECHANICS & INFORMATION ARCHITECTURE (IA)

STAGE 01	STAGE 02	STAGE 03	STAGE 04	STAGE 05
Intent Capture	**Safety Filter**	**Fair Split Math**	**Pass Claim**	**1-Click UPI**
Voice / Hinglish Omnibox NLP	Girls-Only & Boot Volume Gate	Proportional Solo Parity Engine	Live Decrement & Confirmed OTP	NPCI Intent + 4h Ephemeral Purge

2. Screen-by-Screen Functional Wireframes

SCREEN 01 — **Conversational Omnibox**

"IGI Airport T3 kal shaam" REC

IGI Airport (44km) | NDLS Station | Cyber City

Girls-Only Circle: ON | 1 Medium Trolley

UX Mechanics: Single unified search input parses unstructured speech into origin, destination, time, and bag size without multi-field form friction.

SCREEN 02 — **Smart Pool Card & Benchmarks**

Ashwin A. (Host) • 4.9 ★ — 2 Seats Left

Today 06:30 PM • .edu Verified • Girls Safe

Solo Uber: ₹1,400 | Solo Ola: ₹1,350 | **Metro: ₹60**

₹350 (-75% off) — Request to Pool

UX Mechanics: In-card comparison pills transparently show solo market rates, psychologically reinforcing the 75% savings before requesting.

SCREEN 03 — **Verified Boarding Pass**

BOARDING PASS #FS-8924 — CONFIRMED

Pickup: Campus Main Gate 1 (06:25 PM)

Co-Riders: Ashwin A. • Prachi V. • Kaashvi S.

OTP: 8924 — 3/4 Boot Bags

UX Mechanics: Instant state reservation decrements seat counter optimistically and generates a verifiable OTP pass to avoid curb chaos.

SCREEN 04 — **1-Click UPI & Ephemeral Room**

SETTLE FARESHARE: ₹350.00 — 03:59:40

GPay → | PhonePe → | Paytm →

Ephemeral Policy: Chat & location coordinates auto-purge post-trip.

UX Mechanics: Native NPCI universal deep links open UPI apps with pre-filled amount and recipient UPI ID, preventing post-trip payment awkwardness.

3. UI/UX Design System Principles

Zero Cognitive Friction

Campus commuters are in a rush. Replaced traditional 6-step form funnels with an expressive NLP omnibox that understands natural speech and informal abbreviations.

Safety by Default

Female student safety cannot be an afterthought. Built one-tap Girls-Only circles, verified campus identity, and zero-phone-number ephemeral chat drawers.

Tactile Mobile Ergonomics

Designed with 48px+ touch targets, swipeable bottom sheets, high-contrast accessible typography, and instant auditory/haptic feedback on pool confirmation.

FareShare • Architecture & User Journey

End-to-End Functional Capabilities & Engineering Roadmap

VERIFICATION & SCALABILITY

4. End-to-End User Journey (Working Live Today)

STEP 1 Discovery & Smart Filters

Students select a corridor or speak a Hinglish intent. They toggle **Girls-Only Safe Circles** and select their luggage profile. Live matching filters out non-compliant rides instantly.

STEP 2 Solo Fare Cross-Check

Before joining, students click external links to Uber Go, Ola Mini, or DMRC Metro to inspect market baseline pricing with zero sign-in friction, confirming the exact 75% savings ratio.

STEP 3 Booking & Confirmed Pass

Clicking **"Request to Pool"** claims an open seat in real-time state, decrements availability (preventing overbooking), and generates a verified digital Boarding Pass with peer ratings.

STEP 4 Ride Hosting (Host Flow)

If no matching pool exists, students click **"+ Post Ride Pool"**. The algorithmic engine computes proportional fare splits and instantly publishes the ride to the campus feed.

5. User Testing & Feedback Iterations

We validated the prototype through task-based walkthroughs with hostel and day-scholar students commuting to Delhi transit hubs. Their feedback drove three concrete engineering iterations:

🚺 Safety Feedback

Female testers cited random pooling as a blocker. Implemented dedicated **Girls-Only Safe Circles** and verified student credentials.

🎒 Luggage Friction

Airport break travel often overcrowds car boots. Added explicit **Luggage Volume Badges** to prevent mismatched cargo on arrival.

💳 Payment Awkwardness

Testers disliked post-trip Splitwise math. Built **1-Click pre-filled UPI links** (GPay/PhonePe/Paytm) for instant on-the-spot settlement.

6. Production Roadmap: Scaling from MVP to Pan-India

The Big Vision: Moving from preset campus hubs to a universal transit mesh for every university in India.

- **Full Google Maps Platform Integration:** Dynamic pin-to-pin routing anywhere in the city using the *Google Maps Distance Matrix* and *Directions API* for algorithmic detour fare splits.
- **Pan-University Inter-College Mesh:** Expanding from single campuses to inter-college pooling networks connecting neighboring university clusters nationwide.
- **Commercial Dispatch Integration:** Official Uber & Ola Rides API webhooks for automated single-tap ride booking directly from the pool drawer.

🚀 LIVE PRODUCTION REPLIT APP
https://fareshare--pvats2112.replit.app

💻 OPEN-SOURCE GITHUB REPOSITORY
https://github.com/Prachivats-21/fareshare